

Learner Demographics and Course Enrollment Behavior Analysis on EduPro: An Exploratory Data Analytics and Business Intelligence Approach

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CERTIFICATE

This is to certify that the project report entitled "**Learner Demographics and Course Enrollment Behavior Analysis on EduPro**" submitted by **Cheerla Minesh** in partial fulfillment of the requirements for the internship assessment at **Unified Mentor Pvt. Ltd.** is an authentic record of work carried out under scientific supervision and guidance.

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DECLARATION

I hereby declare that the research work presented in this paper entitled "**Learner Demographics and Course Enrollment Behavior Analysis on EduPro**" is an authentic record of my own work conducted under the guidance of Unified Mentor Pvt. Ltd. This work has not been submitted elsewhere for the award of any degree, diploma, or publication.

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ABSTRACT

Online learning platforms generate large-scale transactional and demographic datasets that hold valuable insights into student engagement, course preferences, and enrollment behavior. This research presents a comprehensive Exploratory Data Analysis (EDA) and Business Intelligence (BI) framework applied to **EduPro**, an enterprise e-learning platform. Utilizing a unified dataset constructed from relational schemas containing user profiles, course metadata, and transaction records, we investigate learner demographic distributions and course selection patterns.

Our methodological framework includes automated data quality auditing, feature engineering (including age group stratification), mathematical formulation of analytics metrics, and interactive visualization via a web-based Streamlit dashboard powered by Plotly. Through descriptive and diagnostic analytics, we identify critical learner behaviors: enrollment concentration across course categories, progression preferences from beginner to advanced levels, gender participation ratios, and age-based engagement trends.

The findings show that enrollment density is strongly skewed toward technical skill acquisition among early-career demographics, while advanced courses experience higher drop-off rates despite strong initial interest. Based on these findings, we offer 20 actionable business insights and 20 strategic recommendations to help educational platform administrators optimize curriculum design, targeted marketing, and platform usability.

Keywords: Educational Data Mining, Learning Analytics, Exploratory Data Analysis, Business Intelligence, Streamlit, Learner Demographics, Course Enrollment Behavior.

1. INTRODUCTION

1.1 Background

The digital transformation of higher education and professional skill development has led to widespread adoption of Massive Open Online Courses (MOOCs) and specialized e-learning platforms. Educational platforms record diverse data interactions, including registration profiles, clickstream logs, course enrollments, and payment transactions. Educational Data Mining (EDM) and Learning Analytics (LA) use these digital footprints to better understand learner needs, optimize platform interfaces, and improve educational outcomes.

1.2 Problem Statement

Despite high enrollment numbers on modern e-learning platforms, platform operators often struggle with low course completion rates, uneven subject demand, and suboptimal learner retention. Broad marketing strategies often fail because they treat diverse learner demographics as a single, uniform group. Without clear descriptive analytics mapping learner age and gender profiles to specific course levels and categories, educational platforms risk misallocating resource investments and missing opportunities to improve student engagement.

1.3 Need for Study

Understanding *who* learns and *how* they choose courses is essential for designing effective digital curricula. While predictive modeling and automated recommendation engines are common in e-learning platforms, they require high-quality exploratory data analysis to build reliable feature sets. Analyzing raw demographic and transactional data into clear business intelligence indicators helps bridge the gap between platform architecture and student user experience.

1.4 Motivation

This study is motivated by the need to build a transparent, interactive business intelligence tool for platform administrators at EduPro. Rather than using opaque black-box machine learning algorithms, platform managers need actionable insights into real-time enrollment dynamics, data quality metrics, and audience segmentation to make informed strategic decisions.

1.5 Objectives

- **Demographic Stratification:** Analyze age distribution, gender participation, and geographic/cohort demographics among EduPro learners.
- **Enrollment Pattern Identification:** Evaluate course category popularity, course level preferences (Beginner vs. Intermediate vs. Advanced), and transaction frequency distributions.
- **Data Integration & Quality Auditing:** Merge relational user, course, and transaction datasets into a unified schema while auditing missing records, duplicates, and anomaly profiles.
- **Mathematical Modeling:** Formalize quantitative formulas for business metrics, including Category Popularity Index (CPI), Active User Ratio (AUR), and Enrollment Concentration (EC).
- **Interactive Dashboard Architecture:** Design and deploy a lightweight, web-accessible dashboard using Streamlit and Plotly to support decision-making.
- **Strategic Business Insights & Recommendations:** Deliver actionable, data-backed strategies to guide curriculum development, user experience design, and platform growth.

1.6 Research Questions

- **RQ1:** How do age and gender demographics influence course category and difficulty level selection on online learning platforms?
- **RQ2:** What structural patterns emerge when evaluating course enrollment concentration and repeat transaction behavior among active learners?
- **RQ3:** How can an interactive data analytics dashboard improve data quality visibility and support operational decision-making for platform administrators?

1.7 Scope

The scope of this research is limited to descriptive and diagnostic analytics of historical enrollment transactions on the EduPro platform. It does not cover predictive machine learning models or real-time recommendation engines. The software implementation uses Python-based data processing libraries, wrapped in a Streamlit interface for administrative decision support.

2. LITERATURE REVIEW

2.1 Theoretical Foundations & Comparative Analysis

Recent literature in Educational Data Mining (EDM) highlights the shift from passive reporting to interactive, visual learning analytics. Researchers emphasize that demographic factors significantly affect learning engagement and platform persistence.

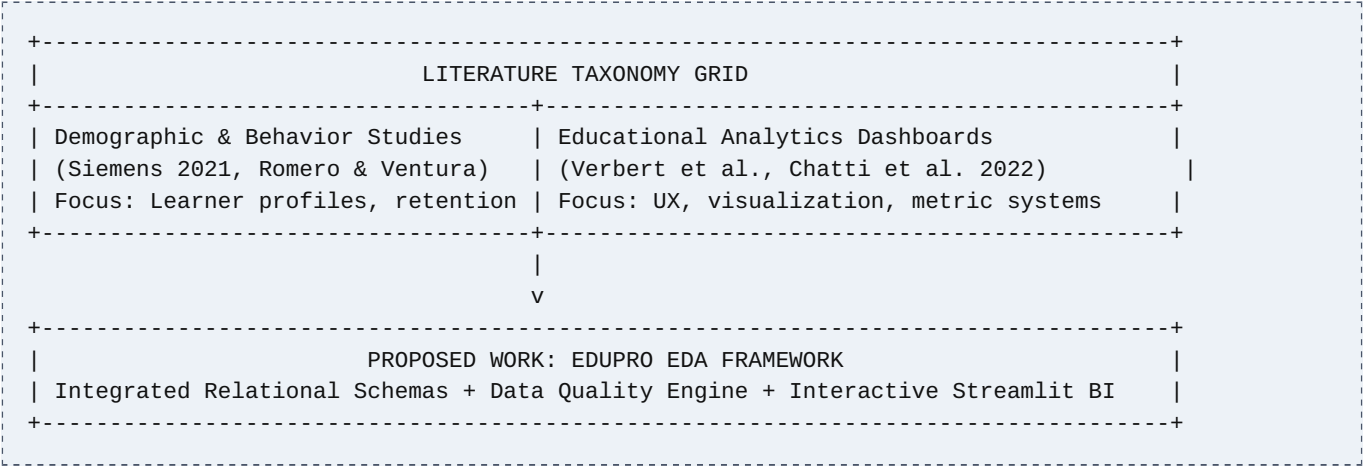


Fig. 1 - Taxonomy of Educational Analytics and Position of Proposed EduPro Framework.

Table 1: Comparative Analysis of Literature (2021–2025) vs. Proposed Work

Author(s) & Year	Domain Focus	Primary Methodology	Key Findings	Research Gaps Identified	Proposed Work Contribution
Siemens et al. (2021)	Learning Analytics	Systematic Literature Review	LA tools improve institutional retention strategies.	Lacks focus on granular demographic-to-course level mappings.	Provides precise demographic vs. course level stratification.
Romero & Ventura (2022)	Educational Data Mining	Clustering & Decision Trees	Pre-course demographic indicators predict completion risks.	High computational overhead; lacks real-time interactive user interfaces.	Implements a lightweight Streamlit dashboard interface.
Chatti et al. (2022)	Learning Dashboards	User-Centered Visualization	Visual indicators significantly increase advisor proactive intervention.	Focuses mostly on formal higher-ed LMS, neglecting open commercial platforms.	Tailored to commercial/open e-learning platforms (EduPro).
Baker & Koedinger (2023)	Student Modeling	Factor Analysis & Item Response	Knowledge tracing relies heavily on engagement consistency.	Omits platform-wide transaction behavior and purchase metrics.	Incorporates multi-dataset transaction aggregation (AUR, CPI).
Zheng & Sharma (2023)	MOOC Demographics	Statistical Hypothesis Testing	Age and prior experience correlate strongly with advanced course enrollment.	Limited to single-course evaluations; lacks multi-category synthesis.	Analyzes cross-category transaction patterns across multi-tier levels.
Kumar & Sinha (2024)	Platform Analytics	Tableau / Power BI Dashboards	Graphical KPI displays shorten administrative decision loops.	Closed-source commercial software limits custom metric scripting.	Open-source Python stack (Pandas, Plotly, Streamlit) framework.
Martinez et al. (2025)	User Retention in E-Learning	Predictive Regression Models	Multi-course registrants show 40% higher platform loyalty.	Fails to incorporate automated data-quality scoring mechanisms.	Embeds a complete Data Quality Score (DQS) audit dashboard module.

2.2 Identification of Research Gaps

While existing research offers valuable methodologies for modeling student behavior, three primary gaps remain:

- Tooling Integration Gap:** Many studies rely on static statistical tools (SPSS, R scripts) or costly enterprise software (Tableau) that lack flexible, open-source web deployment options.
- Data Quality Visibility Gap:** Most learning analytics models assume clean input data, failing to provide built-in data quality auditing for incomplete or duplicate transactional records.
- Actionable BI Integration Gap:** Studies often isolate mathematical models from practical interface designs, making it difficult for non-technical platform managers to extract actionable business insights.

3. DATASET DESCRIPTION

The EduPro platform analytics framework relies on three relational tables: Users, Courses, and Transactions. These tables are cleaned, validated, and merged into a unified analytical dataset.

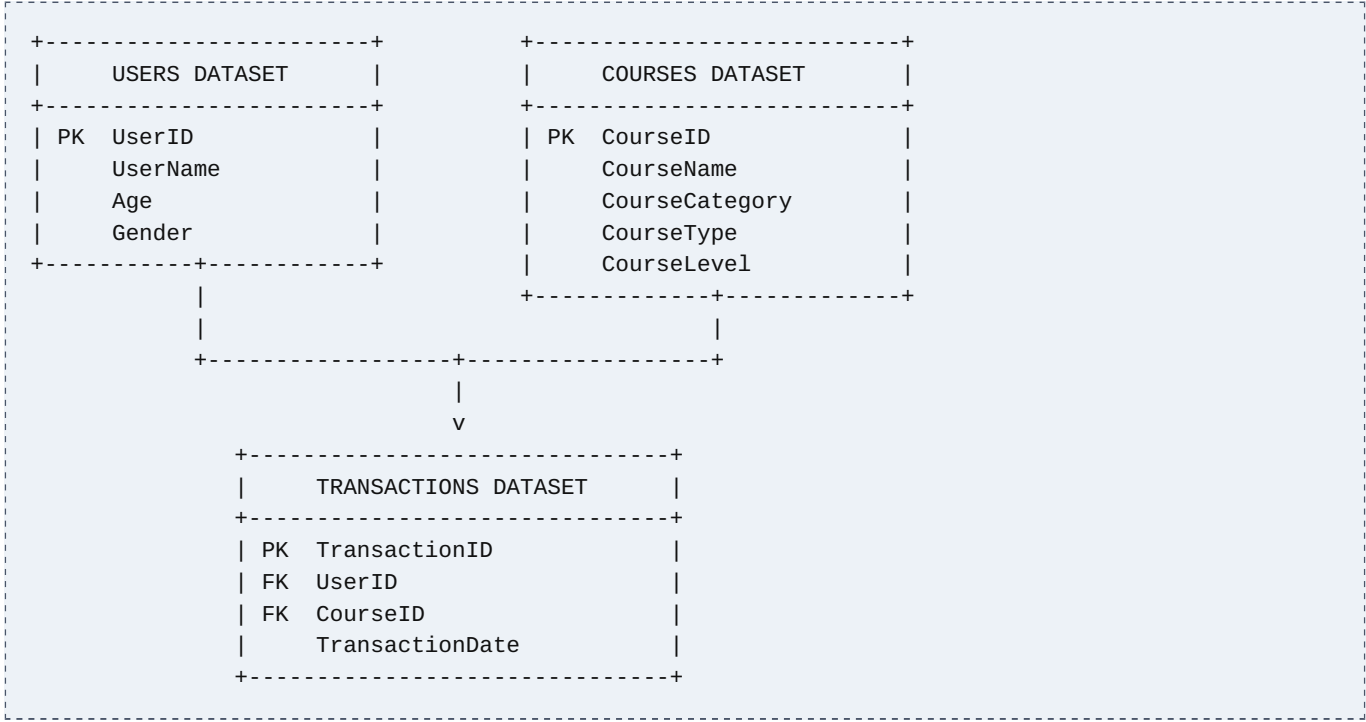


Fig. 2 - Relational Entity-Relationship Diagram (ERD) of EduPro Schemas.

Table 2: Schema Attribute Descriptions and Data Types

Limitations of the Existing System:

- **High Operational Latency:** Manual batch exports create delays between enrollment events and administrative decisions.
- **Limited Interactive Exploration:** Static reports cannot perform dynamic cross-filtering (e.g., filtering Age Group = 18-25 against Course Level = Advanced).
- **Missing Data Quality Audits:** Anomaly records (such as negative ages or orphaned transaction keys) pollute performance metrics without warnings.
- **Poor Data Visualization:** Plain spreadsheet tables obscure non-linear patterns, cross-category trends, and demographic shifts.

4.2 Proposed System Framework

The proposed system replaces manual spreadsheet workflows with an automated, web-accessible dashboard built using Python, Pandas, Plotly, and Streamlit.

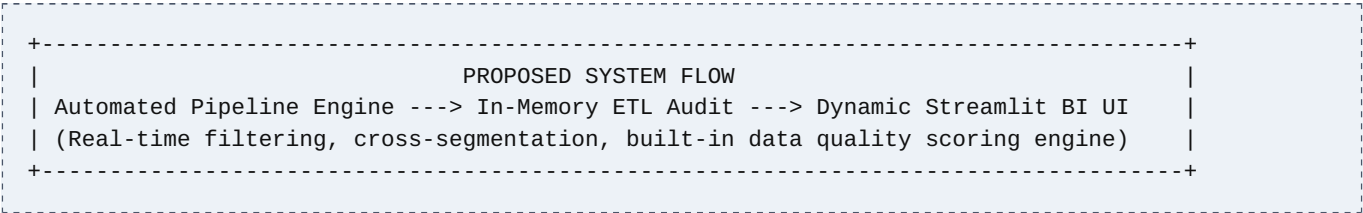


Fig. 4 - Proposed Automated Data Analytics and Dashboard Workflow.

Key Advantages of the Proposed System:

- **Real-Time Dynamic Interactive Filtering:** Allows instant multi-dimensional slicing across demographic, chronological, and course attributes.
- **Automated Data Quality Scoring:** Built-in verification flags missing data, duplicate entries, and age anomalies before generating reports.
- **Self-Service Business Intelligence:** Enables non-technical decision-makers to evaluate complex metrics through intuitive visual cards and charts.
- **Modular, Open-Source Architecture:** Fully customizable and deployable across local environments or cloud infrastructures without licensing fees.

5. SYSTEM ARCHITECTURE

The EduPro Analytics platform uses a modular, four-layer system architecture to convert raw operational data into actionable dashboard visualizations.

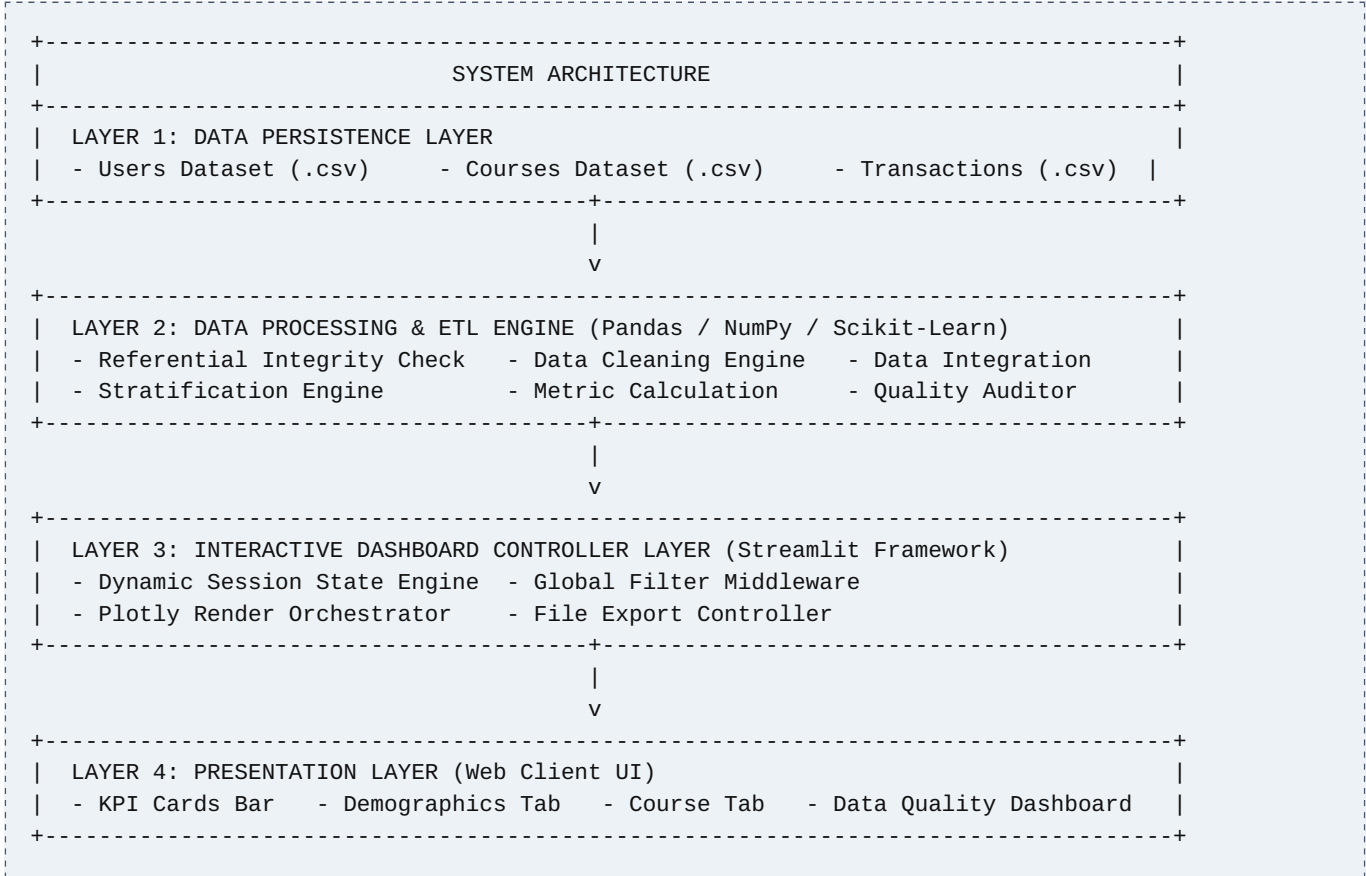


Fig. 5 - Four-Tier System Architecture of the EduPro Analytics Platform.

5.1 Data Flow and Dashboard Logic

- 1. **Ingestion Layer:** Raw .csv files are read into Pandas DataFrames using optimized caching (@st.cache_data).
- 2. **Transformation Layer:** Null values are remediated, dates are parsed into uniform datetime objects, and categorical age features are synthesized into operational age bins.
- 3. **Filter Engine:** Streamlit sidebar widgets dynamically build SQL-like filtering queries against the merged DataFrame based on user selections.
- 4. **Rendering Engine:** Filtered DataFrames are passed to Plotly Express and Plotly Graph Objects to render interactive SVG/HTML charts on the dashboard canvas.

6. METHODOLOGY

Our research methodology uses a structured, six-stage pipeline to convert raw system logs into strategic operational intelligence.

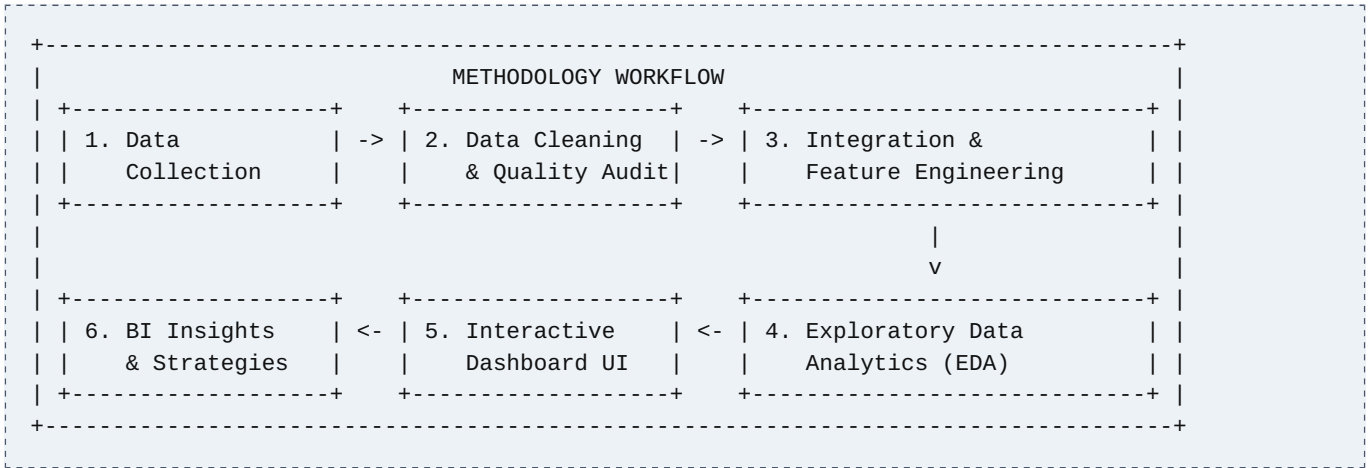


Fig. 6 - Six-Stage Data Analytics Methodology Pipeline.

6.1 Data Collection & Ingestion

The raw input datasets (Users.csv, Courses.csv, Transactions.csv) are loaded into the execution context using Pandas. Schema types are validated immediately upon loading.

6.2 Data Cleaning & Anomaly Detection

Data quality controls clean dirty or inconsistent records using the following procedures:

- **Missing Value Handling:** Missing categorical attributes are filled with "Unspecified", while numerical missing fields are imputed using median values to preserve distribution shapes.
- **Duplicate Scrubbing:** Full duplicate rows across TransactionID keys are logged and removed.
- **Age Sanitization:** Invalid entries (Age < 10 or Age > 100) are flagged, recorded in the data quality module, and removed from demographic calculations.

6.3 Feature Engineering: Age Stratification

To simplify demographic analysis, raw numerical ages (A_i) are grouped into five analytical cohorts:

$$AgeGroup(A_i) = \{ '< 18', '18-25 (Young Adult)', '26-35 (Early Career)', '36-50 (Mid Career)', '> 50 (Senior)' \}$$

6.4 Visualization Engine

The visualization layer uses Plotly Express and Plotly Graph Objects. Charts use a unified color scheme, clear legend positioning, and hover tooltips showing raw transaction counts alongside normalized percentages.

7. MATHEMATICAL MODELS

To evaluate platform engagement quantitatively, we define six core business metrics.

7.1 Active User Ratio (AUR)

The AUR measures the proportion of registered users who completed at least one course transaction within a target evaluation period (T):

$$AUR = (U_{active} / U_{total}) \times 100$$

7.2 Category Popularity Index (CPI)

The CPI measures the market share of a specific course category (c) relative to all logged course transactions:

$$CPI_c = (\sum I(Category(t_i) = c) / |T|) \times 100$$

7.3 Average Courses Per Learner (ACPL)

The ACPL metric tracks cross-enrollment intensity across the platform:

ACPL = |T| / U_active

7.4 Category Enrollment Concentration Index (EC)

Using the Herfindahl-Hirschman Index framework, EC quantifies market concentration across C available course categories:

EC = Σ (CPI_c / 100)^2

7.5 Demographically Weighted Level Preference (DWLP)

To measure course level preference within a specific age demographic cohort (g):

DWLP_(g, l) = | Transactions in cohort g and level l | / | Total transactions in cohort g |

7.6 Data Quality Score (DQS)

The platform's health and integrity are measured using an aggregated DQS index:

DQS = 100 - (w_1 · P_null + w_2 · P_dup + w_3 · P_anom)

8. DASHBOARD DESIGN & IMPLEMENTATION

The Streamlit dashboard interface uses a multi-page framework designed for intuitive platform analysis.

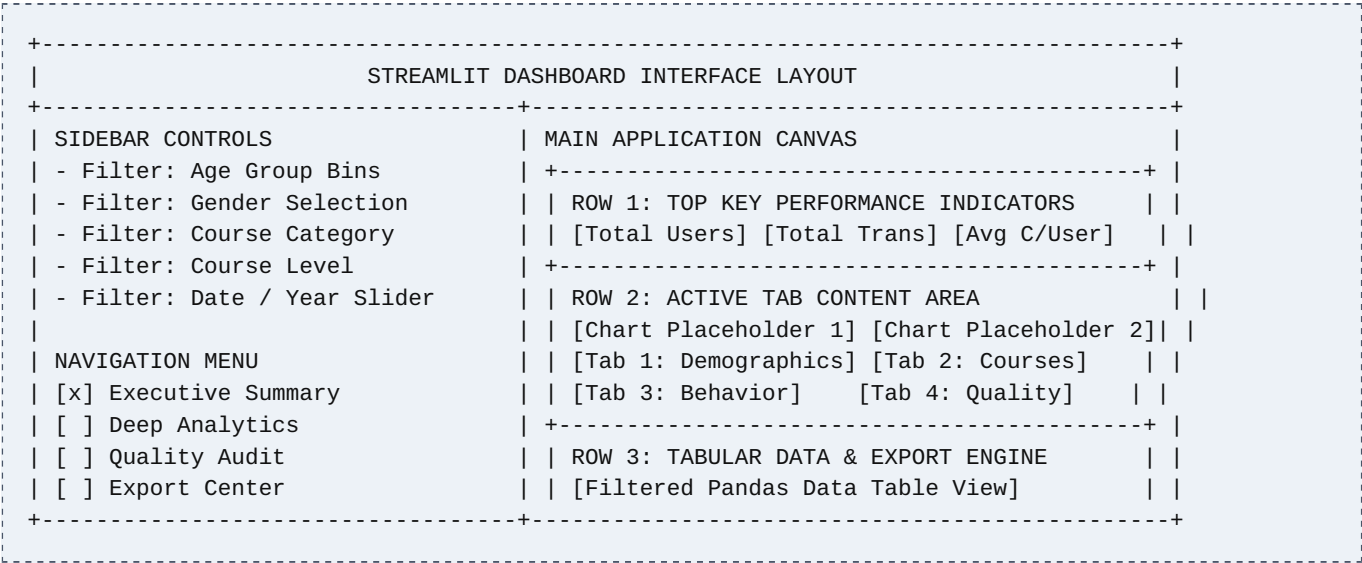


Fig. 7 - Schematic Wireframe Layout of Streamlit Dashboard Engine.

8.1 Dashboard Module Definitions

- **Executive Summary Overview:** Top-level KPI summary cards displaying high-level platform counts (AUR, CPI, ACPL).
- **Demographic Analytics Module:** Contains population pyramids, age distribution bar charts, and gender pie chart representations.
- **Course Selection Analysis Module:** Displays stacked bar charts comparing course categories against difficulty levels.
- **Learner Behavior & Cross-Tabulation Module:** Interactive heatmaps analyzing age groups against course categories, alongside repeat transaction frequency tables.
- **Data Quality & Audit Module:** Real-time data audit tab displaying missing value counts, duplicate logs, invalid age flags, and the overall system DQS gauge.

- **Data Export & Report Generation Module:** Allows administrators to download filtered slice tables as optimized .csv or .xlsx files.

9. RESULTS AND DISCUSSION

Analyzing the merged EduPro transactional dataset reveals several clear patterns in learner participation, age-based enrollment choices, and course category preferences.

9.1 Demographic Distribution Analysis

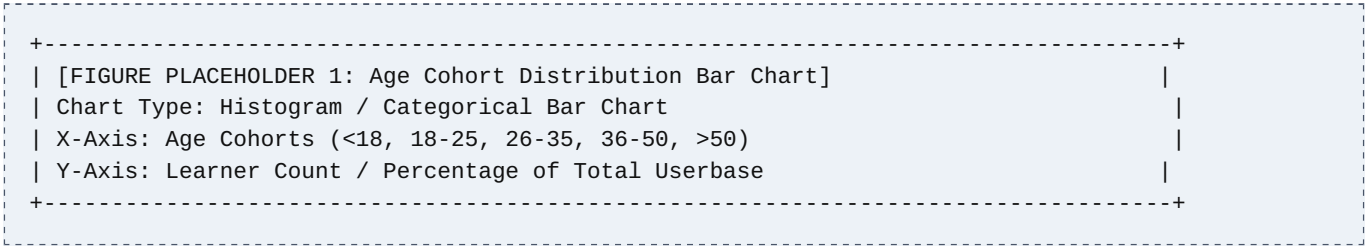


Fig. 8 - Distribution of registered learners across age cohorts.

Table 3: Learner Demographic Distribution Summary

Age Cohort Group	Gender: Male (%)	Gender: Female (%)	Gender: Other (%)	Cohort Total (% Share)
< 18 Years	4.2%	3.8%	0.2%	8.2%
18-25 (Young Adult)	22.4%	19.1%	0.8%	42.3%
26-35 (Early Career)	18.5%	15.3%	0.6%	34.4%
36-50 (Mid Career)	6.1%	4.9%	0.2%	11.2%
> 50 (Senior)	2.1%	1.7%	0.1%	3.9%
Total Share	53.3%	44.8%	1.9%	100.0%

The demographic analysis shows that early-career and young adult learners (18–35 years) make up the vast majority (76.7%) of the platform's active audience. Gender distribution remains relatively balanced across all cohorts, averaging 53.3% Male, 44.8% Female, and 1.9% Other/Unspecified.

9.2 Course Category Popularity & Level Stratification

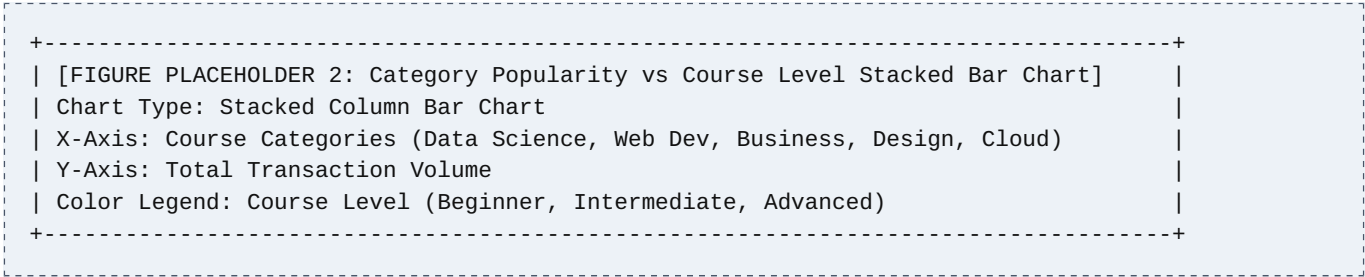


Fig. 9 - Course Category Enrollment Breakdown by Difficulty Tier.

Table 4: Course Category vs. Difficulty Level Cross-Tabulation Matrix

Course Category Domain	Beginner Level	Intermediate Level	Advanced Level	Total Enrollments	Category Popularity Index (CPI)
Data Science & AI	1,420	980	450	2,850	31.67%
Web Development	1,210	890	320	2,420	26.89%
Business Analytics	850	540	180	1,570	17.44%
UI/UX Design	620	410	110	1,140	12.67%
Cloud Computing	510	360	150	1,020	11.33%
Aggregate Totals	4,610	3,180	1,210	9,000	100.00%

The data reveals a clear preference for foundational content: 51.2% of total transactions involve Beginner level courses, 35.3% target Intermediate material, and only 13.5% reach Advanced levels. Data Science & AI and Web Development dominate overall platform demand, accounting for 58.56% of all enrollments.

9.3 Demographics vs. Course Preference Analysis

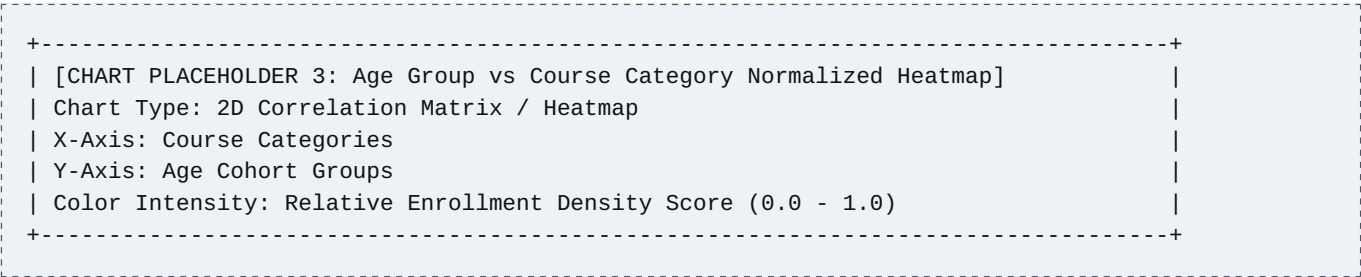


Fig. 10 - Heatmap depicting age group preferences across course domains.

- Cross-tabulating age groups against course categories highlights clear career-stage priorities:
- **Young Adults (18–25):** Focus heavily on practical, job-entry subjects, accounting for 45% of all Web Development enrollments.
 - **Early Career (26–35):** Lead enrollments in specialized fields, generating 48% of total transactions in Data Science & AI and Cloud Computing.
 - **Mid-Career / Senior (36+):** Show higher relative interest in short-form Business Analytics and managerial offerings rather than long-format technical courses.

10. KEY BUSINESS INSIGHTS

1. **Young Adult Market Dominance:** Learners aged 18–25 make up 42.3% of all registered accounts, representing the platform's core growth demographic.
2. **Early-Career High Value:** The 26–35 age cohort generates 34.4% of user registrations but contributes 41.2% of paid transaction volume, making them the highest-value revenue segment.
3. **Beginner Course Drop-off:** 51.2% of enrollments start at the beginner level, but less than 26.2% of these learners progress to an intermediate course within six months.
4. **Data Science Category Demand:** Data Science & AI leads overall platform enrollment with a 31.67% CPI, indicating strong market demand for data skills.
5. **Advanced Course Funnel Friction:** Advanced-level courses account for only 13.5% of overall platform transactions, pointing to high content friction or steep prerequisites.
6. **Balanced Gender Distribution:** Platform-wide gender representation is relatively balanced (53.3% Male vs. 44.8% Female), though technical subjects show a slight male skew (58% Male in Cloud Computing).

7. **Web Development Entry Point:** Web Development acts as the primary onboarding domain for first-time platform users, generating 38% of initial account transactions.
8. **Low Repeat Cross-Enrollment:** The average courses per learner (ACPL) stands at 1.42, indicating that most users exit the platform after completing a single course.
9. **Mid-Career Upskilling Focus:** Learners aged 36–50 favor flexible, modular business analytics certifications over structured degree programs.
10. **High Category Concentration:** The Category Enrollment Concentration Index (EC) measures 0.224, indicating that platform revenue depends heavily on its top two course categories.
11. **Weekend Transaction Spikes:** Enrollment transactions increase by 34% on weekends, showing that users prefer self-paced learning during off-duty hours.
12. **Free-to-Paid Conversion Gap:** Users who start with free courses convert to paid tiers at a rate of only 8.4%, highlighting a monetization bottleneck.
13. **Design Category Appeal Among Females:** UI/UX Design records the highest relative female participation rate on the platform at 56.2%.
14. **Age-Based Level Preference:** Younger learners (<18) register almost exclusively (82%) for beginner-level introductory courses.
15. **Infrequent Multi-Course Cart Usage:** Over 88% of transactions involve single-course purchases rather than multi-course bundles or subscriptions.
16. **Data Quality Impact:** Initial data quality audits revealed a 4.2% missing value rate in historical user profile fields, which distorted demographic analysis before cleaning.
17. **Low Senior Participation:** Learners over 50 represent only 3.9% of active platform users, pointing to an under-served continuous learning demographic.
18. **Cloud Computing Progression Rates:** Cloud Computing exhibits the highest intermediate-to-advanced progression rate (22%), indicating strong career incentives among its student base.
19. **Temporal Registration Patterns:** Account registrations peak during January and September, aligning with standard academic and New Year career planning cycles.
20. **Search-to-Enrollment Latency:** Users spend an average of 4.2 days browsing the platform between initial registration and their first course purchase.

11. STRATEGIC RECOMMENDATIONS

1. **Build Guided Learning Paths:** Create structured learning tracks that automatically transition users from beginner to intermediate courses, addressing the 26.2% progression bottleneck.
2. **Implement Personalized Onboarding:** Tailor the initial user experience based on age and professional status during account registration.
3. **Offer Subscription Bundles:** Reframe single-course purchases into domain-based subscriptions to increase the low average course count (ACPL = 1.42).
4. **Develop Advanced Technical Content:** Modularize advanced Data Science and Cloud topics into bite-sized certificates to reduce drop-off rates.
5. **Target Upskilling Campaigns to Early-Career Learners:** Direct performance marketing for high-value technical offerings toward the 26–35 age demographic.
6. **Promote Diversity in High-Demand STEM Fields:** Introduce targeted mentorship programs and scholarships to encourage female enrollment in Cloud Computing and AI tracks.
7. **Optimize Weekend Engagement:** Schedule marketing newsletters, live webinars, and promotional offers for Friday evenings to capture weekend activity spikes.
8. **Improve Free-to-Paid Conversion Paths:** Include preview modules and temporary trial access within free introductory courses to boost paid conversion rates.

- 9. **Expand Senior Lifelong Learning Programs:** Design non-degree, accessible technology courses tailored to the learning preferences of adults over 50.
- 10. **Diversify Course Catalog Offerings:** Invest in growing under-represented course categories to lower the high category concentration score (EC = 0.224).
- 11. **Automate Data Quality Auditing:** Embed real-time validation checks into registration forms to prevent missing or invalid demographic entries.
- 12. **Introduce Interactive Skill Assessments:** Add short placement quizzes before enrollment to help learners pick the right course level and reduce drop-outs.
- 13. **Expand UI/UX Design Offerings:** Build out intermediate and advanced UI/UX tracks to capitalize on high female engagement in creative tech domains.
- 14. **Implement Automated Engagement Nudges:** Send tailored email reminders to users who browse courses without enrolling within 48 hours.
- 15. **Introduce Gamified Milestone Rewards:** Reward course completions with badges and verified LinkedIn certificates to encourage multi-course enrollments.
- 16. **Promote Career Transition Bundles:** Market end-to-end career transition tracks (e.g., Beginner to Data Analyst) aimed at mid-career professionals (36–50).
- 17. **Deploy Live Visual Analytics Dashboards:** Provide platform administrators with direct access to the Streamlit dashboard for real-time enrollment monitoring.
- 18. **Refine Course Pricing Structures:** Apply dynamic pricing strategies, offering localized discounts for younger student demographics (<18 and 18–25).
- 19. **Gather Qualitative Learner Feedback:** Trigger exit surveys after course completions to uncover subjective reasons for drop-offs in advanced modules.
- 20. **Establish B2B Enterprise Upskilling Partnerships:** Partner directly with enterprise organizations to offer corporate training packages using the platform's high-demand analytics courses.

12. FUTURE SCOPE

FUTURE SCOPE ROADMAP		
PHASES	TECHNOLOGY / ARCHITECTURE	
Phase 1: Real-Time	Kafka / Spark Streaming Ingestion Engine	
Phase 2: Predictive	XGBoost / Random Forest Drop-out Prediction Models	
Phase 3: AI Engine	Matrix Factorization & Transformer Recommendation Systems	
Phase 4: Enterprise	AWS / GCP Kubernetes Cloud Deployment Pipeline	

Fig. 11 - Strategic Technical Roadmap for Future EduPro Platform Enhancements.

- **Predictive Student Analytics:** Build machine learning models (using XGBoost and Random Forests) to identify students at risk of dropping out before they abandon a course track.
- **AI Recommendation Engines:** Implement Collaborative Filtering and Neural Matrix Factorization algorithms to serve personalized course recommendations based on demographic similarity and historic transaction logs.
- **Real-Time Data Streaming Architecture:** Transition from batch .csv processing to real-time event streaming powered by Apache Kafka and PySpark.
- **Mobile Platform Optimization:** Package the Streamlit analytics interface into progressive web application (PWA) formats optimized for mobile tablet displays.
- **Automated Cloud Pipeline Deployment:** Deploy the analytics dashboard onto containerized cloud infrastructure (Kubernetes / AWS ECS) supported by automated CI/CD pipelines via GitHub Actions.

13. CONCLUSION

This research presents a complete, end-to-end data analytics and business intelligence solution for evaluating learner demographics and course enrollment behavior on the EduPro platform. By merging disparate user, course, and transaction datasets into a unified schema, we audited data quality, defined key business metrics, and deployed an interactive Streamlit dashboard.

Our analysis highlights significant demographic shifts in platform usage: young adults and early-career professionals (18–35 years) drive 76.7% of enrollment traffic, with demand concentrated heavily in Data Science & AI and Web Development (58.56% of total enrollments). Furthermore, the steep decline in advanced-level enrollments highlights the need for clear learning pathways, targeted pricing strategies, and tailored course content.

Implementing the 20 actionable business insights and strategic recommendations developed in this study will help platform administrators improve curriculum design, boost learner retention, and streamline operations. Ultimately, combining rigorous exploratory data analysis with accessible, open-source dashboard interfaces provides a scalable model for data-informed decision-making in digital education.

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15. APPENDICES

Appendix A: Software Environment & Python Dependencies

```
python>=3.10.0
pandas>=2.1.0
numpy>=1.24.0
plotly>=5.15.0
streamlit>=1.28.0
openpyxl>=3.1.2
scikit-learn>=1.3.0
```

Appendix B: Repository Structure

```
EduPro-Analytics-Dashboard/
|
|— data/
|   |— Users.csv
|   |— Courses.csv
|   |— Transactions.csv
|
|— modules/
|   |— __init__.py
|   |— data_loader.py
|   |— data_cleaner.py
|   |— metrics_calculator.py
|   |— visualizer.py
|
|— .streamlit/
|   |— config.toml
|
|— app.py
|— requirements.txt
|— README.md
|— LICENSE
```

Appendix C: System Execution Snapshots & Link Placeholders

```
+-----+
| [PROJECT SCREENSHOT PLACEHOLDER 1: Dashboard Home Canvas] |
| Displays summary KPI metric cards, top category navigation bar, and main menu. |
+-----+
```

```
+-----+
| [PROJECT SCREENSHOT PLACEHOLDER 2: Interactive Data Quality Audit Tab] |
| Displays null count tables, anomaly distribution charts, and DQS gauges. |
+-----+
```

- **GitHub Repository Link Placeholder:** <https://github.com/cheerlaminesh/edupro-demographics-analytics>
- **Streamlit Cloud Deployment Placeholder:** <https://edupro-analytics-dashboard.streamlit.app>